MK-UW: Metric Keypoints for Underwater Pose Estimation

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Abstract: Underwater pose estimation remains challenging due to severe image degradation and the scarcity of labeled underwater data. Motivated by the strong performance of metric-keypoint techniques in in-air applications, we propose MK-UW (Metric-Keypoint-based Underwater Pose Estimation), a two-stage transfer learning framework that adapts a geometry-aware metric-keypoint correspondence model from in-air to underwater environments. The framework first performs supervised fine-tuning on synthetic underwater images with preserved ground-truth poses, followed by self-supervised adaptation on real unlabeled underwater images to reduce the remaining domain gap. Experiments on an Aqualoc-based underwater MapFree benchmark demonstrate that MK-UW consistently outperforms direct transfer and representative classical and learning-based baselines.

Keywords: Underwater pose estimation, Transfer learning, Domain adaptation, Self-supervised learning

1 Introduction

Underwater visual pose estimation is important for robotics applications such as navigation, inspection, and mapping [1, 2]. Although learning-based visual pose estimation methods, such as SuperPoint [3] and MicKey [4], achieve strong performance in in-air environments, direct transfer to underwater environments remains unreliable.

The main challenges arise from severe underwater image degradation caused by absorption, scattering, turbidity, and illumination changes [5, 6], together with the scarcity of labelled underwater pose datasets [1]. Consequently, correspondence models trained on in-air datasets often experience substantial performance degradation underwater. Existing underwater pose estimation methods mainly improve feature robustness through image enhancement or appearance-based feature learning [7, 8], while relatively limited attention has been given to transferring geometry-aware correspondence learning frameworks to underwater environments.

To address this problem, this paper proposes metrick keypoints for underwater pose estimation (MK-UW), a two-stage transfer learning framework for adapting geometry-aware metric-keypoint correspondence models from in-air to underwater environments for map-free relative pose estimation without requiring labeled underwater pose datasets. Stage I performs supervised transfer learning on synthetic underwater image pairs generated from labeled RGB-D datasets through underwater image formation models. Stage II further adapts the model to real underwater images using self-supervised learning based on homography consistency and knowledge distillation. In addition, we

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apply an underwater MapFree-style benchmark [9] based on the Aqualoc dataset [10] and demonstrate improved underwater correspondence reliability and pose estimation accuracy over one-stage transfer and representative baselines.

2 Related Work

Correspondence-Based Pose Estimation

Correspondence-based pose estimation recovers camera motion by detecting and matching visual landmarks and solving geometric constraints such as epipolar geometry or PnP [11, 12]. Classical systems such as PTAM [13] and ORB-SLAM [14, 15, 16] rely on traditional feature descriptors including Scale-Invariant Feature Transform (SIFT) [17], Speeded-Up Robust Features (SURF) [18], and Oriented FAST and Rotated BRIEF (ORB) [19]. These methods provide efficient feature matching but often degrade under severe appearance changes.

Learning-based detectors and descriptors such as SuperPoint [20], LF-Net [21], D2-Net [22], R2D2 [23], and DISK [24] improve feature repeatability by learning feature representations directly from training data. Matching networks such as SuperGlue [3] and transformer-based models [25] further improve matching robustness by exploiting global contextual information. More recently, methods such as MicKey [4] jointly learn feature matching and metric pose estimation, making them promising foundations for robust pose estimation under domain shift.

Underwater Visual Pose Estimation

Due to severe underwater image degradation, early underwater SLAM systems adapted classical feature-based pipelines [1, 26, 2], while later methods incorporated image enhancement or physics-based correction before feature extraction [27, 6, 28]. Recent learning-based approaches [7, 8] improve robustness by learning underwater visual features directly from training data. However, these methods remain constrained by limited labelled underwater data and have limited exploration of learning-based feature matching models for underwater pose estimation.

Transfer Learning for Visual Pose Estimation

Transfer learning is a natural solution when labeled target-domain data are scarce [29, 30]. Existing domain adaptation methods range from supervised fine-tuning to self-supervised consistency learning [31]. However, most prior work focuses on image classification or semantic understanding tasks [32, 33], rather than feature matching and pose estimation problems that require geometric consistency across views [34]. Motivated by the strong performance of in-air learning-based feature matching models and the scarcity of labeled underwater pose data, this work investigates how such models can be effectively adapted to underwater visual pose estimation through transfer learning.

3 Method

This section describes the proposed two-stage transfer learning framework for underwater visual pose estimation. The framework progressively adapts a learning-based feature matching model from in-air to underwater environments through supervised transfer on synthetically generated underwater images and self-supervised adaptation on real underwater data. The key objective is to learn geometrically reliable feature correspondences despite severe underwater image degradation.

3.1 Framework Overview

We address the problem of domain adaptation using two types of data: synthetic underwater images with pose annotations and real underwater images without labels. Synthetic underwater images are generated from labeled RGB-D datasets using underwater image formation models [35], preserving geometric relationships between image pairs. The proposed framework adopts a two-stage training strategy. Stage I fine-tunes a pre-trained in-air model on synthetic underwater images using

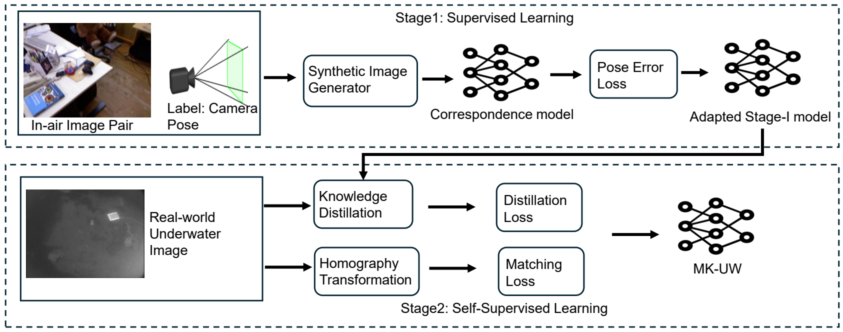

Figure 1: Proposed two-stage metrick-keypoint (MK-UW) transfer framework

geometric supervision, while Stage II further adapts the model on real underwater images using self-supervised learning. The overall framework is illustrated in Fig. 1.

Notation: Let $\mathbb{R}$ and $\mathbb{Z}$ denote the sets of real numbers and integers, respectively. $I \in \mathbb{R}^{H\times W\times C}$ denotes an input image, where $H, W \in \mathbb{Z}$ are the image height and width, and $C \in \mathbb{Z}$ is the number of image channels. A pixel location is represented by $\mathbf{x} \in \Omega$, where $\Omega \subset \mathbb{R}^2$ is the image domain. Let $f_\theta(\cdot)$ denote the feature extraction network parameterized by θ. Given an input image, the network produces a keypoint confidence map $S \in [0,1]^{H\times W}$ and a dense descriptor map $D \in \mathbb{R}^{H\times W\times d}$, where $d \in \mathbb{Z}$ is the descriptor dimension. Given a homography transformation $\mathbf{H} \in \mathbb{R}^{3\times 3}$, the warping operation applied to images or feature maps is denoted by $\mathcal{W}(\cdot, \mathbf{H})$.

3.2 Stage I: Supervised Learning Using Synthetic Underwater Images

Synthetic Underwater Image Generation In Stage I, supervised learning is performed using synthetic underwater images generated from in-air RGB-D datasets with labeled camera poses. The objective is to adapt the pretrained correspondence model to underwater appearance degradation while preserving the geometric constraints required for pose estimation.

Following the physics-based underwater image formation model in [35], the observed underwater image is formulated as

$$\mathbf{U}_c(\mathbf{x}) = \mathbf{I}_c(\mathbf{x})t_c(\mathbf{x}) + \mathbf{B}_c(\mathbf{x})(1 - t_c(\mathbf{x})) + \eta_c(\mathbf{x}), \tag{1}$$

where $c \in \{R, G, B\}$ denotes the color channel, $\mathbf{U}_c(\mathbf{x})$ denotes the observed underwater image, $\mathbf{I}_c(\mathbf{x})$ is the corresponding latent clear in-air image, $\mathbf{B}_c(\mathbf{x})$ represents the background light, and $\eta_c(\mathbf{x})$ denotes additive noise. The transmission term $t_c(\mathbf{x})$ is defined as

$$t_c(\mathbf{x}) = e^{-\beta_c d(\mathbf{x})}, \tag{2}$$

where $d(\mathbf{x})$ is the scene depth at pixel $\mathbf{x}$ and β_c denotes the wavelength-dependent attenuation coefficient for each color channel.

Using this formulation, underwater-style images are generated from in-air RGB-D datasets such as TUM [36] and KITTI [37] while preserving camera pose information. This enables domain adaptation of the correspondence model under reliable geometric supervision derived from known camera poses. Detailed parameter settings and noise configurations are provided in Appendix A.

Once the synthetic underwater dataset is generated, Stage I fine-tunes the pretrained feature matching model under geometric supervision. The proposed two-stage training framework is model-agnostic and does not depend on a specific backbone architecture. Any pretrained feature matching model that predicts matching-related quantities, such as SuperPoint [20] or R2D2 [23], can be adapted in Stage I and further refined in Stage II.

Stage I: Training Using the Generated Synthetic Dataset Once the synthetic underwater dataset is generated, the correspondence model is initialized with pretrained weights using in-air images and fine-tuned using synthetic underwater image pairs with known relative poses. Since the synthetic images preserve geometric annotations from the original RGB-D datasets, supervised training can be directly applied under reliable geometric constraints. The training objective encourages pose-informative correspondences rather than only adapting to underwater appearance.

Given a pair of synthetic underwater images I and I' with ground-truth relative pose T_{GT}, the correspondence model f_θ predicts a set of correspondences J between the two images. These correspondences are then passed to a pose solver $h(\cdot)$, which estimates the relative pose. Stage I loss is defnied as

$$\mathcal{L}_{\text{stage1}} = \lambda_{\text{pose}}\mathcal{L}_{\text{pose}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}. \tag{3}$$

Here, $\mathcal{L}_{\text{pose}}$ penalizes the difference between the estimated relative pose and the ground-truth relative pose T_{GT}, while $\mathcal{L}_{\text{reg}}$ regularizes the predicted correspondences to maintain reliable and geometrically consistent matching. The weights λ_{pose} and λ_{reg} balance pose accuracy and correspondence reliability. The concrete definitions used in our implementation are provided in Sec. 4.1.

After Stage I training, the model adapts to underwater appearance degradation while preserving geometric consistency for pose estimation. This corresponds to the upper half of Fig. 1.

3.3 Stage II: Self-supervised Learning Using Real-world Underwater Images

Although Stage I adapts the model to underwater appearance using synthetic data, a synthetic-to-real gap remains. To improve real-world generalization, Stage II further refines the model using unlabeled underwater images in a self-supervised manner. The Stage I trained model is used as the teacher model in Stage II to preserve the geometry-aware correspondence knowledge learned from synthetic supervised transfer.

Geometric consistency across multiple views is used as supervision without requiring pose annotations. Homography transformations enforce consistent keypoint detection [20], while descriptor consistency preserves reliable feature representations learned in Stage I.

Stage II: Self-supervised Adaptation The proposed self-supervised adaptation stage consists of four complementary objectives: a detection consistency loss $\mathcal{L}_{\text{det}}$ that encourages repeatable keypoint detection under homography transformations, a descriptor consistency loss $\mathcal{L}_{\text{desc}}$ that preserves descriptor similarity for corresponding points while separating non-corresponding features, a sparsity regularization term $\mathcal{L}_{\text{spar}}$ that prevents excessive keypoint activation, and a confidence regularization term $\mathcal{L}_{\text{ent}}$ that encourages stable and discriminative keypoint responses. The overall Stage II objective is defined as

$$\mathcal{L}_{\text{stage2}} = \lambda_1\mathcal{L}_{\text{det}} + \lambda_2\mathcal{L}_{\text{desc}} + \lambda_3\mathcal{L}_{\text{spar}} + \lambda_4\mathcal{L}_{\text{ent}}, \tag{4}$$

where $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are non-negative weighting coefficients. Together, these objectives encourage geometrically consistent, discriminative, and spatially stable feature representations without requiring ground-truth pose annotations. Detailed Stage II equations and implementation settings are provided in Sec. 4. Through this self-supervised adaptation process, the model becomes more robust to real underwater appearance variations while preserving reliable feature matching for pose estimation. This process corresponds to the lower half of Fig. 1.

4 Implementation

This section presents the implementation details of the proposed two-stage training framework, including dataset preparation, training objectives, and self-supervised adaptation strategies.

4.1 Stage I: Supervised Training

For Stage I, synthetic underwater data are generated from the TUM RGB-D dataset [36]. Sequential image pairs with sufficient viewpoint variation are sampled and reorganized into a map-free format, enabling supervised training of pairwise relative pose estimation with geometric consistency.

Teacher Model Selection: We pick MicKey [4] as the teacher model for Stage I adaptation due to its strong in-air pose estimation performance and differentiable correspondence-to-pose pipeline, which is well aligned with the objective of our correspondence-based underwater pose estimation. MicKey is trained on the MapFree dataset [9] with four output heads (detection, descriptor, depth and confidence). Detailed architecture definitions are provided in Appendix B. Based on this specific teacher model, we can construct our detailed loss function.

Stage I Loss: The Stage I objective is defined in (3), where the two terms are

$$\mathcal{L}_{\text{pose}} = \mathbb{E}_{J\sim P(J)}\Big[\ell_{\text{pose}}\big(h(J), T_{\text{GT}}\big)\Big], \tag{5}$$

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{J\sim P(J)}\Big[\ell_{\text{reg}}^{\text{corr}}(J)\Big]. \tag{6}$$

Here, $\ell_{\text{pose}}(\cdot)$ measures the pose estimation error relative to the ground-truth pose T_{GT}, while $\ell_{\text{reg}}^{\text{corr}}(\cdot)$ regularizes correspondence quality, for example through confidence calibration and matching consistency [3, 38]. Detailed definitions and implementation details are provided in Appendix C.

For a candidate feature pair $(\mathbf{x}_i, \mathbf{x}_j)$ with $\mathbf{x}_i \in I$ and $\mathbf{x}_j \in I'$, the correspondence probability is

$$P(\mathbf{x}_i, \mathbf{x}_j) = P_I(\mathbf{x}_i) \cdot P_{I\to I'}(\mathbf{x}_j \mid \mathbf{x}_i) \cdot P_{I'}(\mathbf{x}_j) \cdot P_{I\leftarrow I'}(\mathbf{x}_i \mid \mathbf{x}_j),$$

where P_I and $P_{I'}$ are keypoint selection probabilities, and $P_{I\to I'}$ and $P_{I\leftarrow I'}$ are descriptor-based matching probabilities in the forward and backward directions. A procrustes-based solver [39] with RANSAC is used to estimate the relative pose.

4.2 Stage II: Self-supervised Adaptation

For self-supervised learning, we use the Sea-Thru dataset [6] as a real-world underwater dataset containing diverse scenes with severe visual degradation. Since accurate pose annotations are unavailable, Stage II adopts a self-supervised strategy based on geometric consistency across transformed views, enabling further adaptation to real underwater appearance without labelled pose data.

Stage II Loss:

Stage II training follows the homography-consistency and descriptor distillation framework introduced in Sec. 3.3. Given an input image I, a random homography $\mathbf{H}$ generates a warped image

$$I' = \mathcal{W}(I, \mathbf{H}), \tag{7}$$

where $\mathcal{W}(\cdot, \mathbf{H})$ denotes geometric warping. Since warping may introduce invalid image regions, we define a binary validity mask $M(\mathbf{x})$ and the corresponding valid spatial set

$$\Omega_M = \{\mathbf{x} \mid M(\mathbf{x}) = 1\}. \tag{8}$$

Let $f(\cdot)$ denote the feature extraction network that predicts a detection map and descriptor map. For the original and warped images,

$$(S, D) = f(I), \quad (S', D') = f(I'). \tag{9}$$

Warping the predictions from the original image gives

$$\tilde{S} = \mathcal{W}(S, \mathbf{H}), \quad \tilde{D} = \mathcal{W}(D, \mathbf{H}). \tag{10}$$

The detection consistency term encourages repeatable keypoint detection under geometric transformations, similar to SuperPoint [20],

$$\mathcal{L}_{\text{det}} = \frac{1}{|\Omega_M|} \sum_{\mathbf{x}\in\Omega_M} \Big| S'(\mathbf{x}) - \tilde{S}(\mathbf{x}) \Big|, \tag{11}$$

where $|\Omega_M|$ denotes the number of valid spatial locations. For descriptor consistency, high-confidence locations are selected as

$$\Omega_\gamma = \{\mathbf{x} \mid S(\mathbf{x}) > \gamma\}, \tag{12}$$

where $\gamma > 0$ is the confidence threshold. Descriptor learning is optimized using a contrastive objective inspired by SuperGlue [3] and MoCo [40],

$$\mathcal{L}_{\text{desc}} = -\frac{1}{|\Omega_\gamma|} \sum_{\mathbf{x}\in\Omega_\gamma} \log \frac{\exp\left(\text{sim}(D'(\mathbf{x}), \tilde{D}(\mathbf{x}))/\tau\right)}{\sum_{\mathbf{y}\in\mathcal{N}^-(\mathbf{x})} \exp\left(\text{sim}(D'(\mathbf{x}), \tilde{D}(\mathbf{y}))/\tau\right)}, \tag{13}$$

where $\mathcal{N}^-(\mathbf{x})$ denotes the negative sample set and $\tau > 0$ is the temperature parameter. To regularize keypoint distribution and confidence, sparsity and entropy regularization terms are introduced:

$$\mathcal{L}_{\text{spar}} = \left(\frac{1}{|\Omega|} \sum_{\mathbf{x}\in\Omega} S(\mathbf{x}) - \rho\right)^2, \tag{14}$$

$$\mathcal{L}_{\text{ent}} = -\frac{1}{|\Omega|} \sum_{\mathbf{x}\in\Omega} \Big[S(\mathbf{x}) \log S(\mathbf{x}) + (1 - S(\mathbf{x})) \log(1 - S(\mathbf{x}))\Big], \tag{15}$$

where ρ controls the desired keypoint density. The final Stage II objective is defined in Eq. (4). The settings of $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are provided in Appendix F, while detailed training pipelines for both stages are provided in Appendix D.

For clarity, the above losses are defined for a single image pair. During training, the objective is averaged over a mini-batch B of real underwater image pairs:

$$\mathcal{L}_{\text{Stage II}} = \frac{1}{B} \sum_{i=1}^{B} \left(\lambda_1 \mathcal{L}_{\text{det}}^{(i)} + \lambda_2 \mathcal{L}_{\text{desc}}^{(i)} + \lambda_3 \mathcal{L}_{\text{spar}}^{(i)} + \lambda_4 \mathcal{L}_{\text{ent}}^{(i)}\right), \tag{16}$$

5 Experiment Results

This section evaluates the proposed MK-UW through qualitative comparisons, quantitative benchmarks, and ablation studies against both classical and learning-based baselines. Additional details are provided in the Appendices.

5.1 Qualitative Results

This subsection qualitatively compares feature matching robustness and correspondence consistency under severe underwater image degradation. The comparisons are conducted on multiple underwater datasets, including the grayscale Aqualoc dataset [10] and the RGB Sea-Thru dataset [6]. The Sea-Thru dataset is used only for evaluation and is not included during training. Representative examples are shown in Fig. 2. From left to right, the compared methods are ORB [19], Scale-Invariant Feature Transform (SIFT) [17], SuperPoint [20], UFEN [7], the original MicKey model [4], and MK-UW.

Fig. 2 compares feature matching performance under challenging underwater conditions. Classical methods such as ORB and SIFT often produce many spatially scattered and geometrically inconsistent matches, which become unstable under blur, scattering, low contrast, and illumination variation. Learning-based methods such as SuperPoint and UFEN improve robustness, but many correspondences are still distributed over visually ambiguous or low-texture regions, limiting pose reliability. In contrast, the proposed MK-UW generates fewer but more reliable correspondences concentrated on stable object-level structures such as boundaries, corners, and textured regions. Although sparse, these matches provide stronger geometric constraints for relative pose estimation. This behavior originates from the metric-keypoint formulation inherited from MicKey, which prioritizes pose-informative correspondences. Through the proposed underwater transfer learning framework, MK-UW preserves this correspondence selection behavior while adapting to underwater appearance degradation, resulting in more stable matching and improved pose estimation performance.

5.2 Quantitative Results

This subsection quantitatively evaluates the proposed MK-UW framework on underwater map-free relative pose estimation. The objective is to compare the proposed method against classical, CNN-based, transformer-based, and geometry-aware baselines under challenging underwater conditions.

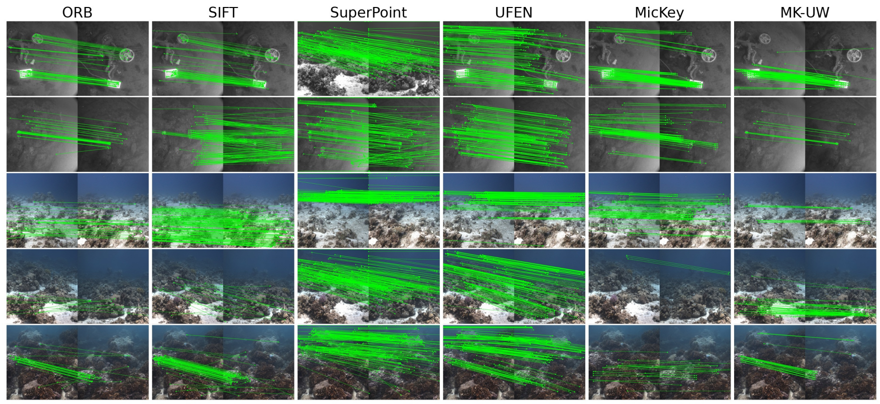

Figure 2: Qualitative comparison of feature matching results on challenging underwater scenes from the Aqualoc [10] (row 1 and 2) and Sea-Thru [6] (row 3 - 5) datasets.

Table 1 compares MK-UW with classical and learning-based baselines on an underwater MapFree benchmark [9] constructed from the Aqualoc dataset [10]. For ORB, SIFT, SURF, SuperPoint, and UFEN, brute-force matching followed by RANSAC is used for relative pose estimation. For DINO-v2, correspondences are obtained using patch-level descriptor similarity before RANSAC. Detailed metric definitions are provided in Appendix E.

Table 1: Results of Transfered Mickey on Underwater Mapfree

Model (steps)	TranE↓	RotE↓	RepE↓	PrecP↑	AucP↑	PrecV↑	AucV↑
ORB[19]	2.41	27.72	194.42	8.41	20.42	21.77	20.70
SIFT[41]	2.70	29.48	207.73	9.09	19.96	20.45	21.01
SURF[42]	3.52	38.54	231.51	8.16	18.74	18.59	17.42
SuperPoint[20]	2.88	24.47	178.71	10.05	20.51	20.68	22.66
UFEN[7]	2.82	23.10	175.42	9.87	21.93	23.74	23.52
DINO-v2[43]	2.65	26.83	180.45	9.13	21.03	22.53	24.58
MicKey (in-air)	1.42	18.80	171.01	9.97	25.35	29.38	46.42
MK-UW (ours)	**1.35**	**15.06**	**142.65**	**13.92**	**29.30**	**37.58**	**57.92**

Table 1 shows that MK-UW achieves the best performance across all evaluation metrics. Compared with the original MicKey model, the proposed framework reduces translation, rotation, and reprojection errors while improving pose precision and VCRE success rates, demonstrating improved robustness under underwater domain shift. Compared with classical descriptors (ORB, SIFT, SURF) and learning-based methods (SuperPoint, UFEN, DINO-v2), MK-UW achieves lower geometric errors and higher pose success rates under underwater blur, scattering, illumination variation, and low-contrast conditions. These results demonstrate the effectiveness of the proposed two-stage transfer learning framework for underwater map-free pose estimation.

5.3 Ablation Study

This subsection evaluates the contribution of different stages and training objectives in the proposed MK-UW framework. The complete pipeline consists of two stages: *Stage I*, which performs supervised transfer learning on synthetic underwater image pairs with ground-truth poses, and *Stage II*, which performs self-supervised adaptation using real underwater images. Table 2 compares different Stage I/Stage II combinations and analyzes the influence of individual Stage II objectives. The first row, *S1+S2 (MK-UW)*, corresponds to the final MK-UW configuration used in the previous experiments. Detailed Stage II components, including homography adaptation, knowledge distillation, and sparsity/entropy regularization, are illustrated in the training pipeline shown in Appendix D.

Table 2: Ablation study on Aqualoc-based underwater map-free evaluation.

Model	TranE↓	RotE↓	RepE↓	PrecP↑	AucP↑	PrecV↑	AucV↑
S1+S2 (MK-UW)	**1.35**	15.06	**142.65**	**13.92**	**29.30**	**37.58**	**57.92**
S1+S2 (All heads)	1.37	15.20	148.92	13.50	28.78	37.28	56.45
S1+S2 (Homo only)	1.38	14.94	159.74	13.60	24.41	36.24	47.11
S1+S2 (KD only)	1.37	**14.63**	160.87	13.08	25.68	36.24	48.39
S1+S2 (No Spar & Ent)	1.38	14.80	157.89	13.50	25.26	35.93	47.45
S2 only (No S1)	1.44	20.01	175.68	12.25	32.91	30.42	57.18
S1 only (No S2)	1.38	15.28	150.83	13.29	26.54	33.30	56.13
MicKey (in-air)	1.42	18.80	171.01	9.97	25.35	29.38	46.42

S1: Stage I ; S2: Stage II; MK-UW: knowledge distillation applied only to the detection and descriptor head; All heads: knowledge distillation applied to all MicKey output heads, including detection, descriptor, depth, and confidence heads; Homo: homography adaptation; KD: knowledge distillation; Spar: sparsity loss $\mathcal{L}_{\text{spar}}$; Ent: entropy regularization loss $\mathcal{L}_{\text{ent}}$; In-air: the pretrained MicKey model.

Table 2 shows that *S1+S2 (MK-UW)* achieves the best overall performance. This demonstrates that combining supervised synthetic transfer with self-supervised real-world adaptation provides the most effective underwater transfer strategy. Compared with *S1+S2 (All heads)*, restricting training on the detection and descriptor head yields better performance, because these heads are directly responsible for producing repeatable keypoints and discriminative local descriptors, which are the most critical components for correspondence establishment. In contrast, applying distillation to all heads may over-constrain the student model. Both *S1+S2 (Homo only)* and *S1+S2 (KD only)* improve over the original MicKey model but remain weaker than the full framework (two-stage MK-UW), indicating complementary benefits from homography consistency and descriptor distillation. Removing sparsity and entropy regularization further reduces performance. Finally, *S2 only (No S1)* produces substantially larger geometric errors, showing that supervised synthetic transfer provides an important initialization for stable underwater adaptation.

6 Limitations

Several limitations of the proposed MK-UW framework should be noted. First, the current framework focuses on pairwise relative pose estimation and does not explicitly model temporal consistency required for visual odometry or SLAM systems. Second, the synthetic underwater image generation process relies on simplified underwater image formation assumptions and may not fully capture complex real underwater phenomena such as dynamic lighting, severe turbidity, marine particles, and moving objects. Third, the current evaluation is mainly conducted on the Aqualoc-based underwater MapFree benchmark, and broader validation on more diverse underwater environments is still required. In addition, the Stage II homography-based self-supervision may become less reliable under strong non-planar scene deformation or large viewpoint changes. Finally, as MK-UW inherits a transformer-based correspondence model, it may require more computation and memory than lightweight hand-crafted features, which could limit deployment on resource-constrained underwater robots.

Future work will investigate underwater visual odometry, multi-view geometric optimization, more realistic underwater simulation, and broader cross-environment adaptation.

7 Conclusions and Discussions

This paper proposed MK-UW, a two-stage transfer learning framework for adapting metric-keypoint correspondence models from in-air to underwater environments for map-free relative pose estimation. Stage I performs supervised transfer on synthetic underwater image pairs, while Stage II applies self-supervised adaptation on real underwater images using homography consistency and descriptor distillation. Experiments on the Aqualoc-based underwater MapFree benchmark show that MK-UW improves correspondence reliability and pose estimation accuracy over direct transfer, classical descriptors, and learning-based baselines. Ablation studies further demonstrate the complementary benefits of supervised synthetic transfer and self-supervised real-world adaptation.

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References

[1] R. M. Eustice, O. Pizarro, and H. Singh. Visually augmented navigation for autonomous underwater vehicles. 2008.

[2] A. Kim and R. Eustice. Pose-graph visual slam with geometric model selection for autonomous underwater ship hull inspection. In *2009 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1559–1565. IEEE, 2009.

[3] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich. Superglue: Learning feature matching with graph neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4938–4947, 2020.

[4] A. Barroso-Laguna, S. Munukutla, V. A. Prisacariu, and E. Brachmann. Matching 2d images in 3d: Metric relative pose from metric correspondences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4852–4863, 2024.

[5] J. S. Jaffe. Computer modeling and the design of optimal underwater imaging systems. *IEEE journal of oceanic engineering*, 15(2):101–111, 2002.

[6] D. Akkaynak and T. Treibitz. Sea-thru: A method for removing water from underwater images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1682–1691, 2019.

[7] J. Yang, M. Gong, and Y. Pu. Knowledge distillation for underwater feature extraction and matching via gan-synthesized images. *IEEE Robotics and Automation Letters*, 2025.

[8] Z. A. Barhoum, Y. Maalla, S. Daher, A. Topolnitskii, J. Mahmoud, and S. Kolyubin. Ur-mvo: Robust monocular visual odometry for underwater scenarios. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 7624–7631. IEEE, 2025.

[9] E. Arnold, J. Wynn, S. Vicente, G. Garcia-Hernando, A. Monszpart, V. Prisacariu, D. Turmukhambetov, and E. Brachmann. Map-free visual relocalization: Metric pose relative to a single image. In *European Conference on Computer Vision*, pages 690–708. Springer, 2022.

[10] M. Ferrera, V. Creuze, J. Moras, and P. Trouvé-Peloux. Aqualoc: An underwater dataset for visual–inertial–pressure localization. *The International Journal of Robotics Research*, 38(14): 1549–1559, 2019.

[11] S. Pan and X. Wang. A survey on perspective-n-point problem. In *2021 40th Chinese control conference (CCC)*, pages 2396–2401. IEEE, 2021.

[12] Z. Zhang. Determining the epipolar geometry and its uncertainty: A review. *International journal of computer vision*, 27(2):161–195, 1998.

[13] G. Klein and D. Murray. Parallel tracking and mapping for small ar workspaces. In *2007 6th IEEE and ACM international symposium on mixed and augmented reality*, pages 225–234. IEEE, 2007.

[14] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardos. Orb-slam: A versatile and accurate monocular slam system. *IEEE transactions on robotics*, 31(5):1147–1163, 2015.

[15] R. Mur-Artal and J. D. Tardós. Orb-slam2: An open-source slam system for monocular, stereo, and rgb-d cameras. *IEEE transactions on robotics*, 33(5):1255–1262, 2017.

[16] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. Montiel, and J. D. Tardós. Orb-slam3: An accurate open-source library for visual, visual–inertial, and multimap slam. *IEEE transactions on robotics*, 37(6):1874–1890, 2021.

[17] T. Lindeberg. Scale invariant feature transform. 2012.

[18] H. Bay, A. Ess, T. Tuytelaars, and L. Van Gool. Speeded-up robust features (surf). *Computer vision and image understanding*, 110(3):346–359, 2008.

[19] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski. Orb: An efficient alternative to sift or surf. In *2011 International conference on computer vision*, pages 2564–2571. Ieee, 2011.

[20] D. DeTone, T. Malisiewicz, and A. Rabinovich. Superpoint: Self-supervised interest point detection and description. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 224–236, 2018.

[21] Y. Ono, E. Trulls, P. Fua, and K. M. Yi. Lf-net: Learning local features from images. *Advances in neural information processing systems*, 31, 2018.

[22] M. Dusmanu, I. Rocco, T. Pajdla, M. Pollefeys, J. Sivic, A. Torii, and T. Sattler. D2-net: A trainable cnn for joint detection and description of local features. *arXiv preprint arXiv:1905.03561*, 2019.

[23] J. Revaud, C. De Souza, M. Humenberger, and P. Weinzaepfel. R2d2: Reliable and repeatable detector and descriptor. *Advances in neural information processing systems*, 32, 2019.

[24] M. Tyszkiewicz, P. Fua, and E. Trulls. Disk: Learning local features with policy gradient. *Advances in neural information processing systems*, 33:14254–14265, 2020.

[25] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

[26] M. Leonardi, A. Stahl, E. F. Brekke, and M. Ludvigsen. Uvs: underwater visual slam—a robust monocular visual slam system for lifelong underwater operations. *Autonomous Robots*, 47(8):1367–1385, 2023.

[27] S. Ansari. A review on sift and surf for underwater image feature detection and matching. In *2019 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)*, pages 1–4. IEEE, 2019.

[28] J. Li, K. A. Skinner, R. M. Eustice, and M. Johnson-Roberson. Watergan: Unsupervised generative network to enable real-time color correction of monocular underwater images. *IEEE Robotics and Automation letters*, 3(1):387–394, 2017.

[29] S. J. Pan and Q. Yang. A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, 22(10):1345–1359, 2009.

[30] M. Wang and W. Deng. Deep visual domain adaptation: A survey. *Neurocomputing*, 312: 135–153, 2018.

[31] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR, 2020.

[32] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell. Adversarial discriminative domain adaptation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7167–7176, 2017.

[33] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. March, and V. Lempitsky. Domain-adversarial training of neural networks. *Journal of machine learning research*, 17(59):1–35, 2016.

[34] R. Hartley and A. Zisserman. *Multiple view geometry in computer vision*. Cambridge university press, 2003.

[35] D. Akkaynak and T. Treibitz. A revised underwater image formation model. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6723–6732, 2018.

[36] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers. A benchmark for the evaluation of rgb-d slam systems. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pages 573–580. IEEE, 2012.

[37] A. Geiger, P. Lenz, and R. Urtasun. Are we ready for autonomous driving? the kitti vision benchmark suite. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012.

[38] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou. Loftr: Detector-free local feature matching with transformers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8922–8931, 2021.

[39] J. C. Gower. Generalized procrustes analysis. *Psychometrika*, 40(1):33–51, 1975.

[40] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9729–9738, 2020.

[41] J. Wu, Z. Cui, V. S. Sheng, P. Zhao, D. Su, and S. Gong. A comparative study of sift and its variants. *Measurement science review*, 13(3):122, 2013.

[42] H. Bay, T. Tuytelaars, and L. Van Gool. Surf: Speeded up robust features. In *European conference on computer vision*, pages 404–417. Springer, 2006.

[43] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum. Dino: Detr with improved denoising anchor boxes for end-to-end object detection. *arXiv preprint arXiv:2203.03605*, 2022.

[44] N. G. Jerlov. *Optical oceanography*, volume 5. Elsevier, 1968.

[45] M. G. Solonenko and C. D. Mobley. Inherent optical properties of jerlov water types. *Applied optics*, 54(17):5392–5401, 2015.

[46] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International conference on machine learning*, pages 794–803. PMLR, 2018.

A Synthetic Underwater Data Generation Setup

This appendix section gives the detailed implementation of synthetic underwater data generation.

For the underwater image formation in Eq. (1) and (2):

$$\mathbf{U}_c(\mathbf{x}) = \mathbf{I}_c(\mathbf{x})t_c(\mathbf{x}) + \mathbf{B}_c(\mathbf{x})(1 - t_c(\mathbf{x})) + \eta_c(\mathbf{x}) \tag{17}$$

$$t_c(\mathbf{x}) = e^{-\beta_c d(\mathbf{x})}, \tag{18}$$

The background light $\mathbf{B}_c$ is defined as [6]

$$\mathbf{B}_c(\mathbf{x}) = \frac{b(\lambda_c)E(d(\mathbf{x}), \lambda_c)}{\beta_c} \tag{19}$$

where $b(\lambda_c)$ denotes the wavelength-dependent scattering coefficient and $E(d(\mathbf{x}), \lambda_c)$ represents the residual light energy at depth $d(\mathbf{x})$ for wavelength λ_c. The residual energy is modeled as

$$E(d(\mathbf{x}), \lambda_c) = E(0, \lambda_c)e^{-K_d(\lambda_c)d(\mathbf{x})} \tag{20}$$

where $E(0, \lambda_c)$ is the light energy at the water surface, $K_d(\lambda_c) > 0$ is the attenuation coefficient of the medium, and $\lambda_c > 0$ denotes the wavelength corresponding to channel c. In this work, we adopt in-air datasets that provide ground-truth camera poses and aligned depth maps as the source of raw images for synthetic generation. Although our experiments use the TUM RGB-D dataset [36] due to its accurate pose annotations and high-quality depth measurements, the proposed generation framework is not restricted to a specific dataset. It can be applied to any in-air dataset with reliable pose and depth information.

The input to our generation model consists of the raw in-air image, its corresponding depth map, and a selected set of parameters β_c, $E(d, \lambda_c)$, and $K_d(\lambda_c)$. The physical parameters are selected using two complementary strategies: (i) random sampling within physically plausible ranges to simulate diverse underwater conditions [6, 44, 45], and (ii) estimation from real underwater images to match specific environments and reduce the domain gap. The output is a set of underwater-style images accompanied by ground-truth camera poses, enabling controlled evaluation of underwater visual algorithms. In this work we picked the latter one and a generated example is shown in Fig. 3.

Figure 3: Examples of synthetic underwater images.

B MicKey Architecture

MicKey [4] is a learning-based keypoint detection and description network designed for metric relative pose estimation. It is trained on in-air image pairs from the MapFree dataset [9]. As shown in Fig. 4, MicKey takes an image as input and extracts shared visual features through a backbone network. These features are then processed by four prediction heads, each responsible for a keypoint-related quantity.

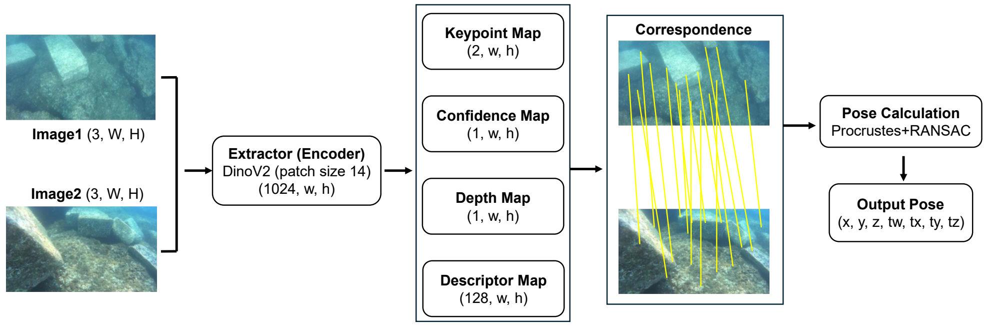

Figure 4: MicKey architecture [4]. The network predicts 2D keypoint offsets $\mathbf{U}$, depth estimates $\mathbf{Z}$, confidence scores $\mathbf{C}$, and 128-dimensional descriptors $\mathbf{D}$, which are used to form probabilistic correspondences and estimate relative pose through a differentiable pose solver.

Specifically, for each image, MicKey predicts 2D keypoint offsets $\mathbf{U}$, depth estimates $\mathbf{Z}$, confidence scores $\mathbf{C}$, and local descriptors $\mathbf{D}$. The 2D offset head $\mathbf{U}$ refines keypoint locations within local image regions. The depth head $\mathbf{Z}$ provides metric depth information associated with predicted keypoints. The confidence head $\mathbf{C}$ estimates the reliability of each predicted keypoint or correspondence candidate. The descriptor head $\mathbf{D}$ outputs 128-dimensional local descriptors used to measure feature similarity across image pairs.

Given two images, the predicted descriptors and confidence scores are used to construct probabilistic correspondences between candidate keypoints. These correspondences, together with the predicted depth and keypoint locations, are passed to a differentiable pose solver to estimate the relative camera pose. This design explicitly links correspondence prediction with metric pose estimation, which makes MicKey suitable as a geometry-aware teacher model for our transfer learning framework.

C Specific loss for stage I

Let the estimated relative pose is $\hat{T} = h(J) = (\hat{\mathbf{R}}, \hat{\mathbf{t}})$, and the ground-truth relative pose is $T_{\text{GT}} = (\mathbf{R}_{\text{GT}}, \mathbf{t}_{\text{GT}})$. Here, $\mathbf{R} \in SO(3)$ denotes a 3×3 relative rotation matrix that describes the orientation change from the source camera frame to the target camera frame, and $\mathbf{t} \in \mathbb{R}^3$ denotes a 3D relative translation vector that describes the displacement between the two camera centers. Accordingly, $\hat{\mathbf{R}}$ and $\hat{\mathbf{t}}$ are the predicted relative rotation and translation, while $\mathbf{R}_{\text{GT}}$ and $\mathbf{t}_{\text{GT}}$ are their ground-truth counterparts.

The specific pose loss and regularization loss from Eq. (5) and (6) is defined as

$$\ell_{\text{pose}}(\hat{T}, T_{\text{GT}}) = \alpha_R d_R(\hat{\mathbf{R}}, \mathbf{R}_{\text{GT}}) + \alpha_t d_t(\hat{\mathbf{t}}, \mathbf{t}_{\text{GT}}), \tag{21}$$

where α_R and α_t balance the rotation and translation terms. The rotation error is measured by the geodesic distance on $SO(3)$:

$$d_R(\hat{\mathbf{R}}, \mathbf{R}_{\text{GT}}) = \arccos\left(\frac{\operatorname{tr}(\hat{\mathbf{R}}^\top \mathbf{R}_{\text{GT}}) - 1}{2}\right), \tag{22}$$

where $\operatorname{tr}(\cdot)$ denotes the trace operator, i.e., the sum of the diagonal elements of a square matrix. This term computes the angular difference between the predicted and ground-truth rotations.

The translation error is defined as the angular difference between the predicted and ground-truth translation directions:

$$d_t(\hat{\mathbf{t}}, \mathbf{t}_{\text{GT}}) = \arccos\left(\frac{\hat{\mathbf{t}}^\top \mathbf{t}_{\text{GT}}}{\|\hat{\mathbf{t}}\|_2 \|\mathbf{t}_{\text{GT}}\|_2}\right). \tag{23}$$

Since monocular relative pose estimation usually recovers translation only up to an unknown scale, this term compares translation directions rather than absolute translation magnitudes.

D Training Pipeline

Fig. 5 illustrates the supervised transfer learning pipeline used in Stage I. Given an in-air image pair (I_1, I_2) with camera poses, we compute the ground-truth relative pose and generate the corresponding synthetic underwater image pair (I'_1, I'_2). The synthetic image pair is then used as input to the fine-tuned model, whose estimated relative pose is supervised by the ground-truth relative pose through the Stage I training loss.

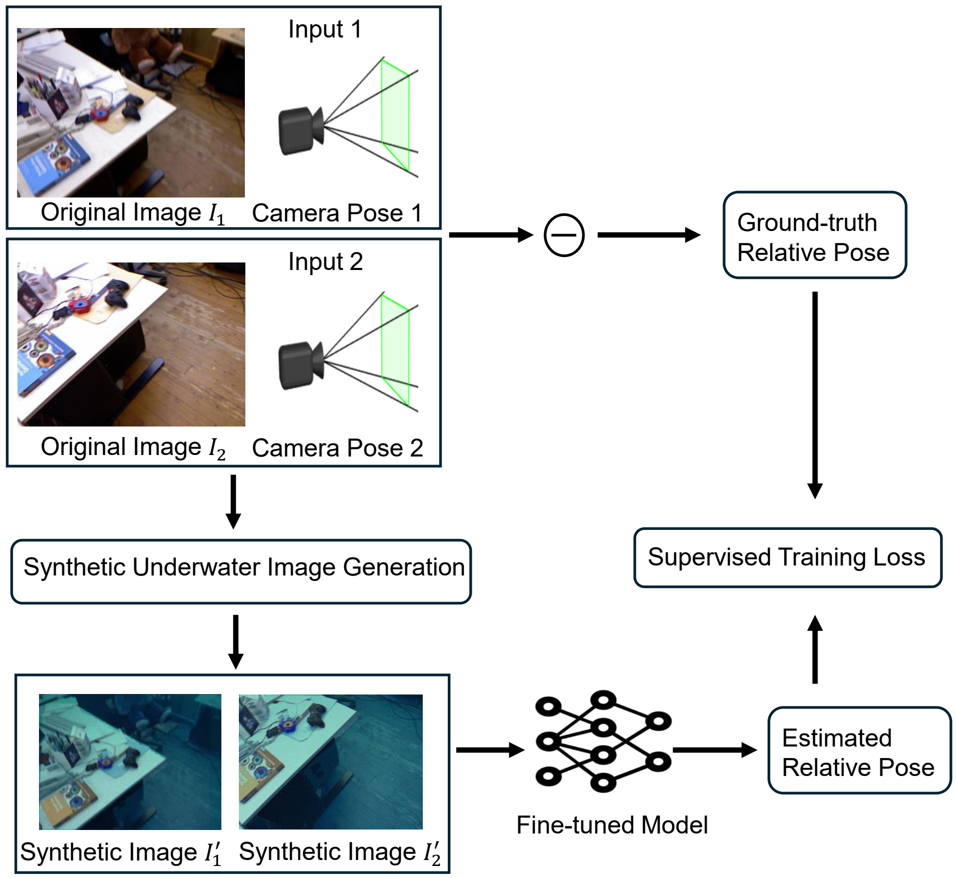

Figure 5: Stage I supervised transfer learning pipeline. In-air image pairs with pose annotations are converted into synthetic underwater image pairs, while the original pose annotations are preserved to supervise relative pose estimation.

Fig. 6 illustrates the self-supervised adaptation pipeline used in Stage II. Given an unlabeled real underwater image, a homography transformation is applied to construct a geometrically related image pair. The student model is trained using feature consistency, descriptor consistency, sparsity, and entropy losses. In addition, knowledge distillation from the teacher model is used to preserve the geometry-aware correspondence representation during adaptation.

E Evaluation Metrics

We follow the map-free relocalization evaluation protocol introduced in [9]. Given a reference image and a query image, the task is to estimate the metric relative pose of the query camera with respect to the reference image. Let the ground-truth query pose be $T = [R \mid \mathbf{t}] \in SE(3)$ and the estimated pose be $\hat{T} = [\hat{R} \mid \hat{\mathbf{t}}] \in SE(3)$, where $R, \hat{R} \in SO(3)$ and $\mathbf{t}, \hat{\mathbf{t}} \in \mathbb{R}^3$. In our evaluation, lower values

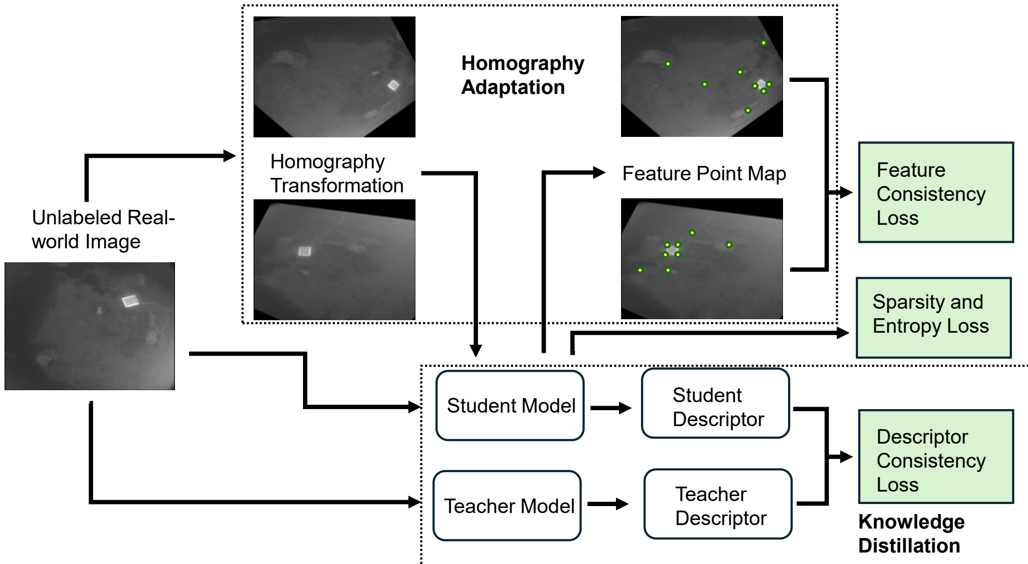

Figure 6: Stage II self-supervised adaptation pipeline. Real underwater images are adapted through homography-based consistency learning and teacher-student knowledge distillation, without requiring ground-truth underwater pose labels.

are better for translation error, rotation error, and reprojection error, while higher values are better for precision and area-under-curve metrics.

Translation error. Translation error measures the Euclidean distance between the estimated and ground-truth camera centers:

$$e_t = \|\hat{\mathbf{c}} - \mathbf{c}\|_2 , \tag{24}$$

where $\mathbf{c}$ and $\hat{\mathbf{c}}$ denote the ground-truth and estimated camera centers in the reference coordinate frame, respectively. When poses are represented in camera-to-world form, the camera center is directly given by the translation vector. When poses are represented in world-to-camera form, the camera center is computed as $\mathbf{c} = -R^\top \mathbf{t}$.

Rotation error. Rotation error measures the angular difference between the estimated and ground-truth rotations:

$$e_R = \cos^{-1}\left(\frac{\mathrm{tr}(\hat{R}R^\top) - 1}{2}\right) \cdot \frac{180}{\pi}. \tag{25}$$

The error is reported in degrees. In implementation, the argument of $\cos^{-1}(\cdot)$ is clipped to $[-1, 1]$ for numerical stability.

Virtual Correspondence Reprojection Error. Following [9], we also report Virtual Correspondence Reprojection Error (VCRE), denoted as e_{vcre}. VCRE measures the image-space misalignment induced by the estimated pose. Since the map-free setting does not assume a reconstructed 3D map, a set of virtual 3D points $\mathcal{P} = \{\mathbf{p}_i\}_{i=1}^N$ is placed in the query camera coordinate system. These points are transformed and projected using both the ground-truth and estimated poses. The VCRE is then defined as the average Euclidean distance between the corresponding projected pixels:

$$e_{\text{vcre}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\pi\left(KT^{-1}\mathbf{p}_i\right) - \pi\left(K\hat{T}^{-1}\mathbf{p}_i\right)\right\|_2 , \tag{26}$$

where K is the camera intrinsic matrix and $\pi(\cdot)$ denotes perspective projection from homogeneous camera coordinates to image coordinates. VCRE is reported in pixels. Intuitively, this metric estimates how much virtual content would be displaced in the image if rendered using the estimated pose instead of the ground-truth pose.

Pose precision. Pose precision evaluates the fraction of image pairs whose predicted pose is within predefined translation and rotation thresholds. A pose estimate is considered correct if

$$e_t < \tau_t \quad \text{and} \quad e_R < \tau_R, \tag{27}$$

where τ_t and τ_R are translation and rotation thresholds, respectively. In the original MapFree benchmark implementation, the default thresholds are $\tau_t = 0.25\text{m}$ and $\tau_R = 5°$. The pose precision is computed as

$$\text{Prec}_P = \frac{1}{M} \sum_{m=1}^{M} \mathbb{I} \left[e_t^{(m)} < \tau_t \wedge e_R^{(m)} < \tau_R \right], \tag{28}$$

where M is the number of evaluated image pairs and $\mathbb{I}[\cdot]$ is the indicator function.

VCRE precision. VCRE precision evaluates the fraction of image pairs whose reprojection error is below a pixel threshold:

$$\text{Prec}_V = \frac{1}{M} \sum_{m=1}^{M} \mathbb{I} \left[e_{\text{vcre}}^{(m)} < \tau_v \right], \tag{29}$$

where τ_v is the VCRE threshold. The MapFree benchmark commonly uses VCRE thresholds corresponding to a percentage of the image diagonal, such as 5% or 10%. In the public evaluation code, a default threshold of 90 pixels is used for the single-frame benchmark.

Confidence-aware AUC. In addition to measuring whether an estimate is correct, the MapFree protocol also evaluates whether a method can assign meaningful confidence to its predictions. Let $s^{(m)}$ be the confidence score of the m-th pose estimate. For a confidence threshold γ, predictions with $s^{(m)} \geq \gamma$ are accepted, while the remaining predictions are rejected. This gives a precision-recall curve:

$$\text{Precision}(\gamma) = \frac{\sum_{m=1}^{M} \mathbb{I}[s^{(m)} \geq \gamma] \mathbb{I}[\text{correct}^{(m)}]}{\sum_{m=1}^{M} \mathbb{I}[s^{(m)} \geq \gamma]}, \tag{30}$$

$$\text{Recall}(\gamma) = \frac{\sum_{m=1}^{M} \mathbb{I}[s^{(m)} \geq \gamma]}{M}. \tag{31}$$

The area under this confidence-based precision-recall curve is reported as AUC. When the correctness condition is defined using Eq. (27), the metric is denoted as AucP. When correctness is defined using $e_{\text{vcre}} < \tau_v$, the metric is denoted as AucV.

Reported metrics. In the main paper, we report the following metrics:

- **TranE**: average median translation error e_t.
- **RotE**: average median rotation error e_R.
- **RepE**: average median reprojection error, corresponding to VCRE in the MapFree protocol.
- **PrecP**: precision under the joint translation and rotation pose threshold.
- **AucP**: confidence-aware AUC under the joint pose threshold.
- **PrecV**: precision under the VCRE threshold.
- **AucV**: confidence-aware AUC under the VCRE threshold.

F Training Parameter Setup

The implemented Stage II objective in equation (4) is

$$\mathcal{L}_{\text{stage2}} = \lambda_1 \mathcal{L}_{\text{det}} + \lambda_2 \mathcal{L}_{\text{desc}} + \lambda_3 \mathcal{L}_{\text{spar}} + \lambda_4 \mathcal{L}_{\text{ent}},$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are non-negative loss weights.

For the Stage II objective in Eq. (4), we select the weighting coefficients $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ using two practical strategies.

Fixed balancing strategy: We first monitor the raw magnitudes of the four sub-losses during warm-up iterations and set fixed coefficients so that their weighted contributions are of similar scale. For example, we observe that the entropy term $\mathcal{L}_{\text{ent}}$ is often around two orders of magnitude larger than $\mathcal{L}_{\text{det}}$ and $\mathcal{L}_{\text{desc}}$. Therefore, we set λ_4 about 100 times smaller than λ_1 and λ_2 (and similarly tune λ_3) to avoid domination by a single term.

Normalized-gradient strategy: As an alternative, we adapt weights according to gradient magnitude inspired by GradNorm [46]. Let $g_k = \|\nabla_\theta(\lambda_k \mathcal{L}_k)\|_2$ for $k \in \{1, 2, 3, 4\}$ and $\bar{g} = \frac{1}{4}\sum_{k=1}^{4} g_k$. We update coefficients to equalize gradient scales, for example

$$\lambda_k \leftarrow \lambda_k \left(\frac{\bar{g}}{g_k + \epsilon}\right)^{\alpha},$$

where $\epsilon > 0$ ensures numerical stability and $\alpha \in (0, 1]$ controls update strength. This strategy reduces manual tuning and keeps all sub-objectives active during optimization.

We have done a sensitivity analysis for these parameters. The default parameters for λ is 0.5, 0.5, 0.01, 0.001.

Table 3: Sensitivity analysis of Stage II hyperparameters on Aqualoc-based underwater map-free evaluation.

Parameter	Value	TranE↓	RotE↓	RepE↓	PreP↑	PreV↑	AucP↑	AucV↑
λ_1 (Det)	0.25	1.380	15.14	160.02	13.3	28.2	36.7	55.0
	1.0	1.375	15.01	159.02	13.3	27.8	36.2	54.5
	2.0	1.378	14.84	159.73	13.2	29.6	35.8	54.2
	4.0	1.378	14.83	159.66	13.2	28.7	35.6	53.3
λ_2 (Desc)	0.25	1.384	15.10	157.55	13.5	27.8	35.5	53.1
	1.0	1.380	14.84	159.32	13.5	28.3	36.1	54.4
	2.0	1.381	14.75	159.86	13.6	26.9	38.4	56.2
	4.0	1.361	15.82	172.28	11.9	26.9	29.9	51.2
λ_3 (Spar)	5e-3	1.380	14.79	159.43	13.1	28.6	31.2	54.7
	0.1	1.379	14.88	158.45	13.3	29.3	29.0	49.6
	1.0	1.375	14.89	160.83	13.5	27.6	24.7	45.1
	2.0	1.377	14.62	157.40	13.2	27.3	24.0	46.0
λ_4 (Ent)	5e-4	1.380	14.76	159.55	13.5	28.5	31.6	53.7
	0.01	1.382	14.49	156.82	13.6	28.6	28.4	48.7
	0.1	1.379	15.24	157.43	13.3	29.3	26.2	49.4
	1.0	1.378	15.15	160.24	13.6	28.0	26.9	49.3